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Synthesis, characterization and optical properties of graphene oxide–polystyrene nanocomposites

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The synthesis of graphene oxide (GO)–polystyrene (PS) Pickering emulsions, as environment-friendly nanostructures suitable for novel applications, has received significant attention in recent years. In this work, the synthesis and characterization of GO–PS nanocomposites through seeded emulsion polymerization and the selective light reflection properties of dry films have been reported. Amphiphilic molecule sulfonated 3-pentadecyl phenol was used as a co-surfactant to stabilize GO dispersions during the emulsion polymerization process. The particle size of the dispersions as measured by dynamic light scattering decreases from 540 nm, for PS without any GO, to 88 nm with 1 wt% GO content. Scanning electron microscopy studies show a uniform size distribution of the composite particles prepared with GO. The dried films show a structural color that varies with the GO content. The self-assembly behavior of the dried film was further studied using reflectance spectroscopy, which shows a red shift of the reflectance maximum from 440 to 538 nm as the GO loading was increased from 0.2 to 0.5 wt%, respectively, indicating a different microstructure. X-ray diffraction, transmission electron microscopy (TEM) and atomic force microscopy (AFM) were used to study the morphology and structure of the composite particles on drying. The AFM study confirms the non-spherical shape of the particles. Thermogravimetric analysis shows improved thermal decomposition characteristics of the nanocomposite films. **Copyright © 2015 John Wiley & Sons, Ltd.**

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Keywords: graphene oxide; Pickering emulsions; nanocomposites; particle size; morphology; optical properties

INTRODUCTION

The discovery of graphene during the last decade has added a new dimension to polymer nanocomposite research. Graphene-based polymer composites are reported to display superior mechanical, thermal, barrier, electrical and flame retardant properties.[1–5] The special properties of graphene, the intermediate graphene oxide (GO) and its derivatives have been widely reported.[2,6–8] In contrast to graphene, GO exhibits better compatibility with organic polymers[9,10] and has received considerable attention as a nanofiller for polymer nanocomposites.[5] It is not surprising that in a short period of time, a number of excellent reviews have appeared focusing on the synthesis, characterization and derivatization or functionalization of graphene.[11–15] The availability of reliable and scaled-up production methods for GO and reduced GO (rGO) offer a wide range of possibilities to synthesize graphene-based functional materials for various applications.[16–18] In recent years, the intermediate GO has been studied widely for its unique hydrophobic–hydrophilic characteristics and optoelectronic properties.[19–23] The aqueous emulsion polymerization method has been widely investigated as an environment-friendly route for the preparation of GO-based nanocomposites.[20–24]

Pickering emulsion polymerization, where the GO acts as stabilizer for the formed particles of styrene, has been one of the most widely studied.[25,26] Stabilization of polystyrene (PS) colloidal particles by GO was studied by Song *et al*.[27] Graphene–PS composites were synthesized via emulsion polymerization by Zhang *et al*.[26] The composites displayed better thermal, mechanical and conductivity properties than pure PS. Functionalized graphene was found to be an effective nanofiller with very good dispersability in the PS matrix. Han *et al*. prepared nanocomposites of GO and graphene with PS and studied the flame retardancy properties.[6] Zhang *et al*. used sulfonated GO as a new and effective material for the synthesis of PS-based nanocomposites via Pickering emulsion polymerization.[18] The effect of pH and initiator type on the Pickering emulsion polymerization of GO-stabilized PS was also reported by Xie *et al*.[28] They obtained wrinkled particles with a water soluble initiator but spherical particles with azobisisobutyronitrile (AIBN) as a monomer soluble initiator and hexadecane as an auxiliary stabilizer. The diameter distribution was narrow, and hollow particles were obtained.

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Mozami and Sharif studied the Pickering emulsion approach for polymerization of methyl methacrylate with AIBN and obtained a broad distribution of particle size varying from 500 nm to about 5 microns.[20] Emulsion polymerization with persulfate initiators has been reported, but there is no information on the particle size characteristics or the product properties.[25] The various reports thus show that the Pickering emulsion approach can be effectively used for the preparation of nanocomposite latexes, but several challenges like control of particle size, size distribution and morphology are of interest when compared with conventional emulsions. Controlling the structure and morphology of the resulting nanocomposites would facilitate the development of high-performance nanocomposites. In this article, we report the synthesis and characterization of a range of GO–PS nanocomposite dispersions in the presence of sulfonated 3-pentadecyl phenol (SPDP) as a new co-surfactant.[29] The study shows that varying the concentration of GO allows control of particle size and its assembly into various nanostructures. The effect of GO content on the optical properties of GO–PS nanocomposites has been reported for the first time.

EXPERIMENTAL

Materials

Flake graphite (carbon content of 99.9%) supplied by Rangoon mill stores, Secunderabad, India was used for the GO synthesis. Potassium permanganate ($KMnO_4$), sodium nitrate ($NaNO_3$), sulfuric acid (H_2SO_4), hydrogen peroxide (H_2O_2) and sodium bicarbonate ($NaHCO_3$) were purchased from SD. Fine Chem., Mumbai and used as such. Styrene was procured from Sigma-Aldrich (India) and distilled under vacuum and stored at −18°C till use. The initiator, potassium peroxodisulfate ($K_2S_2O_8$), was obtained from Aldrich.

Synthesis of GO dispersions and Pickering emulsions

GO was prepared from flake graphite using the modified Hummers method.[30] One gram of graphite powder and 0.5 g of sodium nitrate were added to 70 ml of concentrated H_2SO_4 taken in a two-neck round bottom flask kept in an ice bath. Then, 3 g of $KMnO_4$ were gradually added. The mixture was stirred for 2 h and diluted with deionized (DI) water. Afterwards, 30% H_2O_2 solution was added to the flask until the color changed to bright yellow, indicating oxidation of the graphite. Then, the solution was neutralized. The graphite oxide slurry obtained was structurally characterized and re-dispersed in DI water and exfoliated to generate graphene oxide nanosheets (GONS) by ultrasonication using a bath sonicator for 1 h. To prepare stable dispersions, SPDP was used as the co-surfactant. The resulting aqueous dispersion of GONS was stable. The surfactant concentration was kept constant in all cases at 0.1 mg/ml. The temperature during the sonication was maintained constant at 60°C.

The Pickering emulsion was prepared using the aqueous dispersion of GO as a seed in the emulsion polymerization. Polymerization of the emulsion was carried out using a general recipe for seeded emulsion polymerization. In a typical experiment, 50 g of GO–water dispersion containing the SPDP surfactant was taken in a three-neck round bottom flask. Five milliliter of styrene monomer was added through a syringe and stirred at 250 rpm for 1 h using a mechanical stirrer under N_2 atmosphere. The pH of the solution was maintained at 9.0–10.0 by adding $NaHCO_3$. The reaction mixture was heated to 75 ± 2°C, and 50 mg of $K_2S_2O_8$ was added to initiate the polymerization. The reaction was run for 24 h to achieve the maximum conversion in all cases.

Characterization

The IR spectrum was recorded on a Perkin Elmer machine. The particle size of the GO dispersions as well as the Pickering emulsion was measured using a light scattering machine from Precision Detectors, USA, and the zeta potential was measured with a Malvern Nanosizer from Malvern Instruments, UK.

The Raman spectra were recorded on a Horibo-Jobin-Yvon LABRAM Spectrometer, using an excitation wavelength of 632.8 nm, and the experiments were performed under ambient conditions. Powder X-ray diffraction (XRD) was performed with a SIEMENS X-ray diffractometer employing Cu–Kα radiation ($\lambda = 1.5406$°A) under ambient laboratory conditions.

The FE–SEM analysis was performed on dried specimens after gold sputtering on a HITACHI S-4300SE/IN scanning electron microscope. The internal structure of the particle was studied using transmission electron microscopy (TEM). The TEM analysis was performed on a JEOL JEM 2010F machine at an accelerating voltage of 160 kV. The samples for TEM analysis were prepared by placing a drop of the diluted dispersion on carbon grids, which were then dried.

The reflectance spectrum of the dried film on a glass slide was measured on a CARY 5000 UV-VIS spectrophotometer (model: UV 1205 M011) from Agilent Technologies, USA.

The morphology of the dried film on a freshly cleaved mica surface was studied using atomic force microscopy (AFM). Atomic force microscopy experiments were preformed with a Nanowizard® II instrument from JPK Instruments AG, Germany under tapping mode.

Thermal stability was studied on a TA Q-500 thermogravimetric analyzer. The measurement was conducted under N_2 atmosphere at a heating rate of 10°C/min from 30 to 600 °C.

RESULT AND DISCUSSION

The *in situ* emulsion polymerization method has recently been applied by several groups to prepare GO-polymer composites in aqueous media, and various aspects of the synthesis have been studied.[20–27] The resulting composites are called Pickering emulsions, where the GO functions as a stabilizer for the growing polymer particles.[24–26] In this approach, the surfactant characteristics of GO are utilized and the properties are also pH and size dependent. Because we observed some agglomeration of the GO particles during polymerization, SPDP was used as an auxiliary surfactant. Larger particles of GO have a lower specific surface area resulting in lower GO concentrations adsorbed on the particle surface, which leads to instability and precipitation.[20] In the following sections, we first discuss the synthesis of GO and then the nanocomposite synthesis as well as its properties.

SYNTHESIS AND CHARACTERIZATION OF GO DISPERSIONS

GO was synthesized from graphite using the modified Hummers method.[30] GO possess a large fraction of sp3 hybridized C atoms bound to oxygen functional groups. The IR spectrum of

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pristine graphite and its oxidized form are given in Fig. 1. It shows characteristic peaks at 1629 and 1402 cm^{-1} corresponding to the stretching vibration of C=C in the benzene ring of graphite. The figure also shows vibration bands as a result of hydroxyl groups, carbonyl groups, C–O bond and the epoxy groups centered at 3420, 1702, 1359, 1228 and 1093 cm^{-1} respectively, which were not present in the starting material.

The structure of GO was further confirmed with XRD (Figure S1, supporting information), Raman spectroscopy (Figure S2, supporting information) and elemental analysis. The XRD result of the graphite shows a sharp peak at $2\theta = 26°.5'$ corresponding to diffraction from the (002) plane with a d-spacing of 0.34 nm. In GO, this peak nearly vanishes and a major peak at a lower value ($2\theta = 10°$) was observed with a corresponding d-spacing of 0.87 nm. The increase of the interlayer spacing in GO is attributed to intercalation by functional groups such as epoxy and hydroxyl groups, the surfactant and also by moisture.[31] Analysis by Raman spectroscopy shows an increase of the D/G ratio from 0.23 for flake graphite to 1.5 for GO. The increased D band intensity in the Raman spectra is conventionally attributed to the disorder introduced by the formation of sp3 hybridized carbons. The elemental analysis shows that the C/O ratio increases to 1.3 for GO. These results confirm that highly oxidized GO has been synthesized and agrees well with the earlier reports on GO synthesis.[15]

In the next step, a dispersion of GO in DI water was prepared. Huang *et al.* established that GO can work as a functional surfactant as a result of the presence of hydrophobic graphene clusters and hydrophilic groups similar to those of an amphiphilic block copolymer.[21] Synthesis of GO dispersions with a maximum concentration of 1 mg/ml has been reported earlier. Ruos *et al.* prepared dispersions with higher GO concentrations, up to 5 mg/ml, using polystyrene sulfonate (PSS) as a stabilizer.[32] The surfactants play an important role in stabilizing monolayer or multilayer graphene dispersions, and the obtained dispersions are useful in the preparation of optical and electronic devices.[6] It was necessary to obtain seed dispersions with higher loading of GO and at the same time maintain stability of the composite dispersion at the temperature at which emulsion polymerization was carried out. Because we observed some settling of GO dispersion during the polymerization without surfactant, an auxiliary surfactant was used to obtain stable dispersions. The amphiphilic molecule, SPDP is a low molecular weight co-surfactant, whose surfactant properties have been reported earlier.[29] When polymerized, the GO sheets decorate the polymer particles on the surface and thus prohibit restacking and keep them separate. The particle size of the final emulsion also depends on the dispersion method used, and the presence of a surface active agent was expected to reduce the emulsion particle size.

The determination of particle size using dynamic light scattering shows that the GO particles were nearly monodisperse [cf. Fig. 2, histogram of the prepared GO dispersion, concentration 1 mg/ml]. The particle size varied around an average value of 280 ± 30 nm as the GO content was increased from 0.1 to 1 mg/ml. However, for the lowest GO concentration of 0.1 mg/ml, the average size was 500 nm (plot not shown). The increase of size at the lower GO concentration could be because of the fact that in a dilute dispersion, the sheets may exist in a more extended form and at higher concentrations where folding or coil formation of the sheets might be taking place. The existence of twisted and coiled ultra long multilayer graphene ribbons or wrinkled structures have also been earlier reported.[33–35] While studying the self-assembly of GO on a xylene droplet surface, Mozami and Sharif also observed the ability of GO sheets to fold into bubble-like structures and preserve this morphology even in the dry state.[20] It was suggested as reflecting the extremely high flexibility of GO sheets despite their high modulus. They observed that when the lateral size of the GO sheet was larger than the emulsion droplet size, highly crumbled and wrinkled sheets were observed.

SYNTHESIS AND CHARACTERIZATION OF COMPOSITE DISPERSIONS

Aqueous dispersions of GO with varying solid content were prepared and used as a seed in the emulsion polymerization of styrene using potassium persulfate initiator. The schematic of

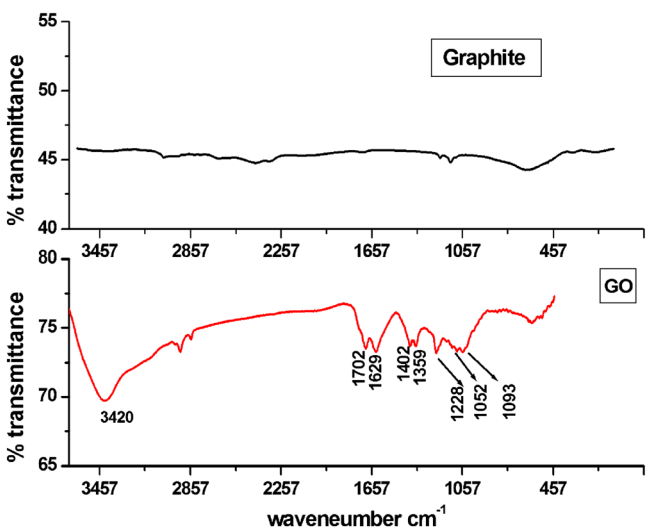

Figure 1. FT-IR spectra of graphite and graphene oxide. This figure is available in colour online at wileyonlinelibrary.com/journal/pat

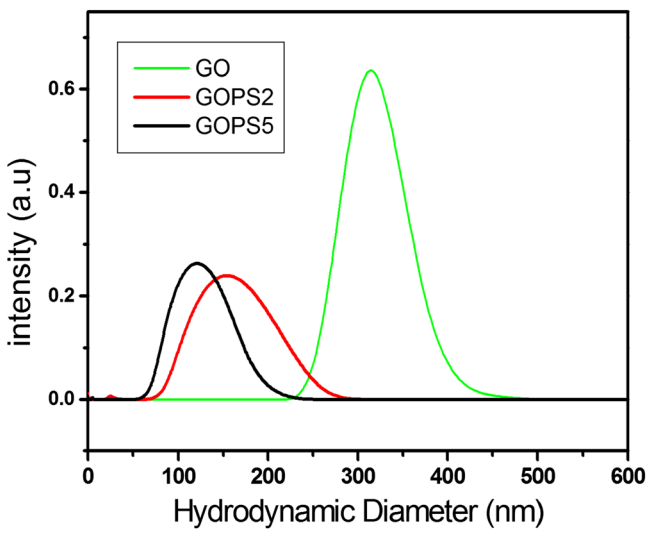

Figure 2. Particle size distribution plots of graphene oxide and nanocomposite dispersions in water obtained from dynamic light scattering. This figure is available in colour online at wileyonlinelibrary.com/journal/pat

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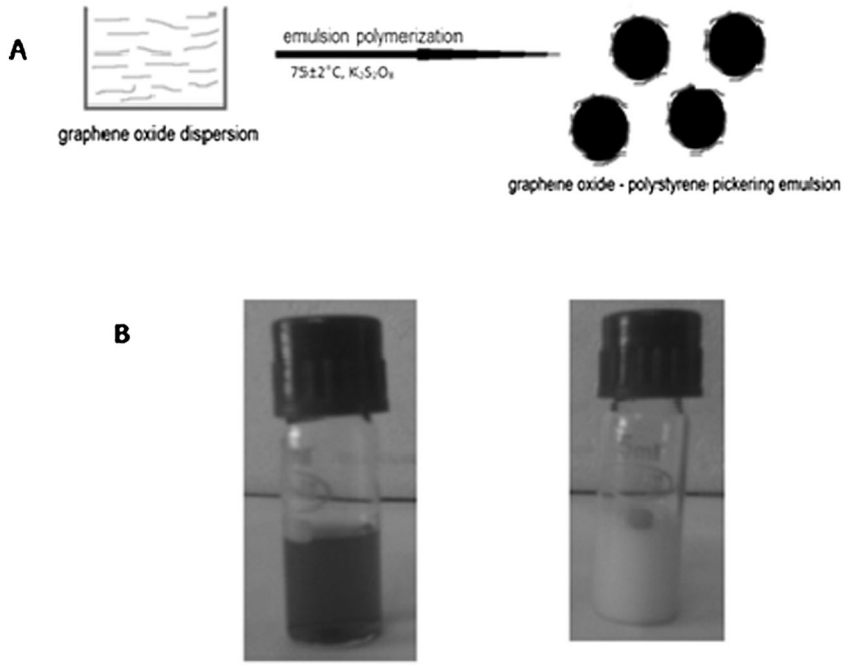

Figure 3. (A) Schematic of the preparation procedure used for graphene oxide–polystyrene (GO–PS) Pickering emulsions and (B) images of graphene oxide dispersion (left) and the prepared Pickering emulsions (right).

the GO–PS Pickering particle formation is depicted in Fig. 3A. The images of the prepared GO dispersion and the resulting Pickering emulsion are shown in Fig. 3B.

The particle size distribution was again monomodal in all cases as seen from Fig. 2. A stable Pickering emulsion was obtained in all cases. The formulations used and the characteristics of the Pickering emulsions prepared with GO are summarized in Table 1. It can be seen that the particle size of the composite dispersions decreases with increasing concentrations of GO. The size decreases from 539 nm for PS without any GO to 88 nm for the Pickering emulsion with 1 wt% GO. According to earlier studies, the hydrophilicity of GO is size dependent[21] and suggests a decrease in surface tension with increasing GO concentrations. The zeta potential values (cf. Table 1) also showed a more negative value after polymerization indicating the enhanced stability of the Pickering emulsions. The GO seed dispersions showed an average zeta potential value of −37 mV, which is in agreement with the values reported in the literature. Earlier studies reported that GO sheets are stable in the aqueous phase as a result of the polar epoxy, carboxyl and hydroxyl groups and the bearing of negative charges under alkaline conditions (zeta potential~−50 mV).[20]

Although the self-assembly behavior of GO at the liquid–liquid interface has been studied by several groups for the preparation of Pickering emulsions, detailed characteristics of the prepared emulsions are not known. Mozzami[20] studied the self-assembly behavior of GO at the methyl methacrylate–water interface to achieve a uniform distribution of graphene in the prepared composites and studied the thermodynamic forces necessary for self-assembly. Mozami found that in polymerizations employing a GO/polymer weight ratio higher than 4 wt%, stable dispersion was obtained. The lack of dispersion stability was based on an insufficient concentration of nanolayers. They suggested that sufficient quantity of nanoparticles at the interface of the polymer particles is crucial for stabilization, and this critical concentration appears to be correlated to the ratio of the specific surface area of the nanoparticles to the resulting polymer particles (S) and also the surface coverage of the polymer particles with the nanoparticles. In addition to the surface area of GO, the polymer particle size and extent of coverage of polymer particles with GO affect the critical concentration of nanolayers for stabilization. In our case, it appears that a synergistic combination of the surfactant and GO may be helping to reduce the particle size.

Table 1. Composition and characteristics of GO–PS Pickering emulsions and the homopolymer emulsion[a]

Code	Graphene oxide[b] (weight %)	Styrene (g)	Conversion[c] (%)	Zeta potential (mV)	Particle dia.[d] (nm)
PS2	0	4.65	70	−28	539
GOPS1	1	4.65	74	−42	88
GOPS2	0.5	4.65	80	−52	150
GOPS5	0.2	4.65	77	−37	125
GOPS4	0.1	4.65	72	−45	292

[a]1 wt% $K_2S_2O_8$ was used as the initiator in all cases.
[b]Seed dispersions have a size of 280 ± 30 nm in all cases and were prepared with 0.1 mg/ml SPDP surfactant.
[c]Determined by gravimetric methods.
[d]Determined from dynamic light scattering. The zeta potential value was determined with a Malvern Nanosizer.

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SCANNING AND TRANSMISSION ELECTRON MICROSCOPY

The morphological characterization of the resulting hybrid colloidal particles was examined to study the arrangement of GO sheets and the polymer particles. Typical SEM images of the dried dispersions are shown in Fig. 4, which exhibits peculiar self-assembly characteristics.

The morphology of the particles changes from spherical close-packed particles with 0.2 wt% GO to elongated ellipsoid structures as the GO content is increased to 0.5 wt%. For PS without any GO, no regular packing was observed and a broad size distribution of the particles can be seen. To study the internal structure of the particles, TEM analysis was also carried out, and the images are shown in Fig. 5. The images further confirm the uniform size of the particles. The image (5B) also shows the surface of the particles covered with the GO sheets as schematically depicted in Fig. 3. This is also in agreement with earlier studies, which suggested a rough surface for the particles as they are covered with the GO sheets.[28,31]

In addition to the structural order, the dried films exhibit an iridescent color under visible light and the color also varies with

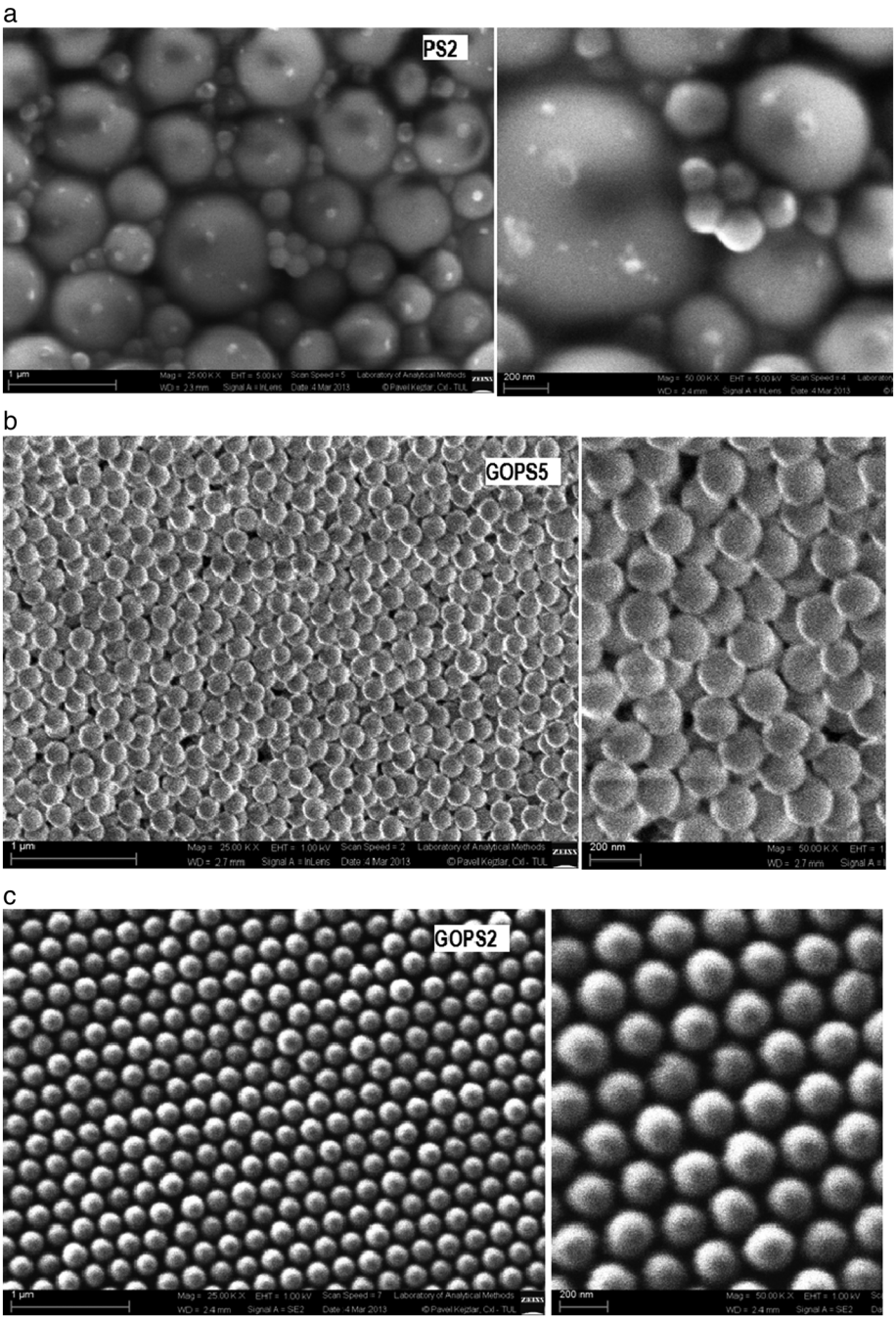

Figure 4. Scanning electron micrographs of graphene oxide–polystyrene Pickering emulsions with varying GO loading. The images show a change in the shape of the particles as well as particle size distribution as a function of the graphene oxide content. a) PS2 without any GO, b) GOPS5 with 0.2 wt% GO, and c) GOPS2 with 0.5 wt% GO.

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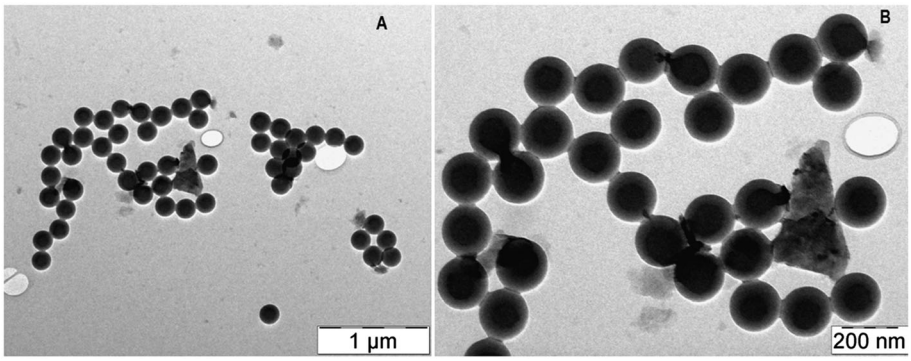

Figure 5. TEM image of GOPS2 showing the internal structure of the Pickering emulsion particles.

the GO content. Although the preparation of GO–PS Pickering emulsions have been reported earlier, their self-assembly into ordered opaline structures is not known and the observation of a structural color in these systems has not been reported to the best of our knowledge. This is shown in Fig. 6A along with the corresponding reflection spectra of GO–PS films.

The existence of "structural color" as a result of the refractive index contrast in the periodic structure of self-assembled colloidal photonic crystals resulting from constructive interference of visible light at specific wavelength has been well studied.[36–38] The required contrast has often been achieved with core-shell particles via interface modification. For example, color tunability was achieved through engineering the chemical composition of the interface in core-shell precursor particles to obtain a larger refractive index (RI) contrast.[36,37] Bo *et al.* have also demonstrated the structural color in colloidal crystal films using a mixture

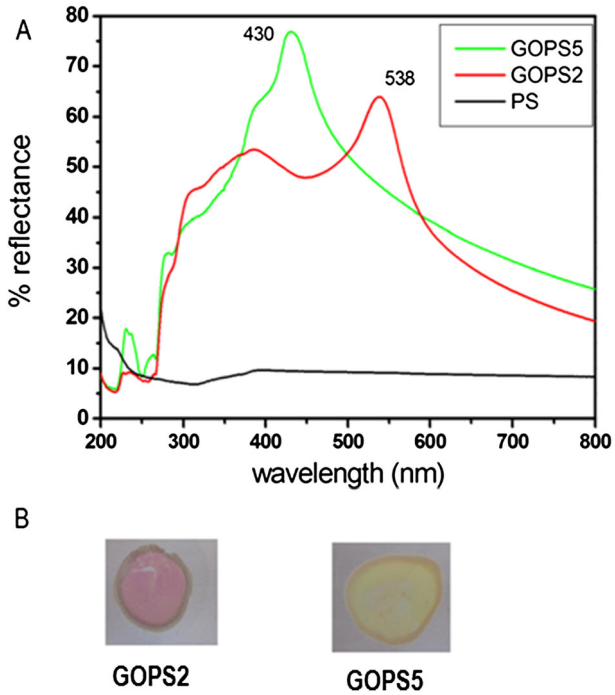

Figure 6. (A) Reflectance spectrum of GO–PS nanocomposite showing variation of the reflectance maximum as a function of the GO content. (B) Optical images show the color of dried films; pink for the sample with 0.5 wt% GO content (GOPS2) and yellow as the GO content is decreased to 0.2 wt% (GOPS5). This figure is available in colour online at wileyonlinelibrary.com/journal/pat

of soft polymer spheres with colloidal silica.[38] Obviously, the peak position of the stop band and the color of the film depend on the refractive index contrast between the polymer and the nanofiller plus air, the relative filling ratio and the diameter of polymer spheres. Accordingly, the photonic bands shifted from 435 to 640 nm, showing that the structural color is tunable.

The reflectance spectra of GO–PS films shown in Fig. 6A along with the color exhibited by the dried films shown in Fig. 6B suggest the formation of self-assembled 3D-ordered structures similar to the colloidal crystal films. The samples prepared with GO content show a reflectance peak, which red shifts from 440 to 538 nm as the GO content increases. Thus, the color can be tuned from colorless to yellow and then to pink as the GO content is increased from 0 to 0.2 wt% and then to 0.5 wt%, respectively. To explain these results, we assume that the Pickering emulsion on drying gives films characterized by a uniform distribution of GO in the PS matrix. The peak wavelength of the stop band and the color of 3D periodic colloidal structures have been conventionally explained using the Bragg diffraction law[38]:

$$\lambda = 2\bar{n}d\sqrt{1 - \frac{1}{n^2}sin^2\theta}$$

where d is the distance between crystalline planes in the (111) direction and related to the polymer sphere diameter D by $d = \sqrt{2/3}\,D$ for *fcc* structure and n is the average refractive index. The color tuning of opaline structures by addition of below 50 nm size carbon particles has been reported earlier.[39] Synthetic opals prepared from core-shell particles with a controlled interface using fluorinated monomers were shown to display structural color as a result of the RI contrast between the two polymers. Graphene is reported to possess a complex high refractive index [3 + 1.5i], and the required RI contrast for achieving the structural color is thus achieved.[40] The photonic band gap theory also predicts that the reflectance is strongly dependent on the ratio of the refractive indices of two materials, i.e. $n_{refl} = n_{high} / n_{low}$.[38] Accordingly, a high index material is required to achieve a complete 3D band gap showing 100% reflectance at the wavelength of interest.

To understand the broad shape of the reflectance peak, interactions between the GO and PS were considered. GO possess an atomic structure with non-uniform coverage of oxygen containing functional groups on the basal plane. In carbon materials containing a mixture of sp2 and sp3 bonding, the optoelectronic properties are reported to be mainly determined by the π and π^* states of the sp2 sites, which lie within the σ–σ^* gap.[41] Because the π bonding is weaker and has a lower formation energy, a

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large number of disorder-induced localized states are present in the 2D network of synthesized GO. The optical transitions between the disorder-induced localized states cause a broad absorption or emission band.[41] These disorder-induced transitions are expected to be further extended through conjugation with the benzene ring of PS and the polar groups of the surfactant. The SEM images confirmed the formation of spherical, uniform-sized particles at lower GO concentrations. In comparison, PS prepared without any GO (PS) possesses a broad particle size distribution (c f. Figure 4a).

ATOMIC FORCE MICROSCOPY

AFM is a widely used technique to study the morphology of latex dispersions. To study the morphology of GO-stabilized Pickering emulsions, the sample was applied on a freshly cleaved mica surface and the height images were recorded using tapping mode AFM. The AFM images of dispersions are given in Fig. 7 and show close packing of the particle in the height image. It can be seen that the 3D images show elongated particles as seen in the FE SEM images (c f. Figure 4c).

Pickering emulsions with non-spherical particles require only lower concentrations for full surface coverage. The adsorption of submicron size GO layers on the surface of the polymer particles can make the appearance of particle edges look rough. The nanolayers fold on the surface of the polymer particles, reducing the interfacial tension between the particles and water, which results in partial coverage of the polymer particles.

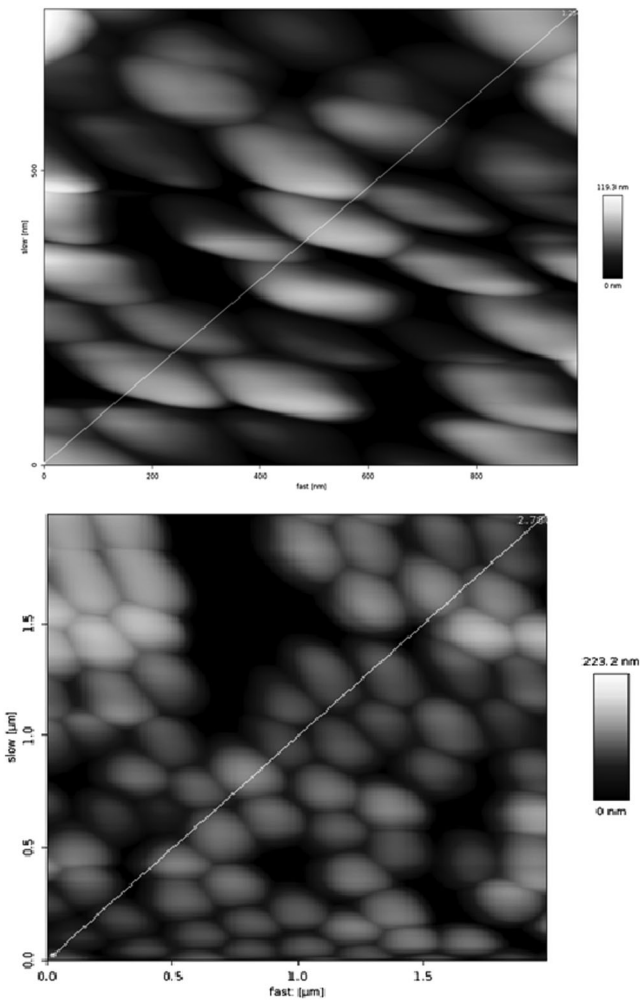

Figure 7. AFM images of the dried dispersion GOPS2 showing morphology of the drying particles at two different magnifications.

XRD OF NANOCOMPOSITES

X-ray diffraction is a widely used technique to characterize nanocomposite structure. The XRD pattern of the homopolymer and that of the nanocomposite are compared in Fig. 8. GO shows a characteristic diffraction peak at 2θ value of 10° corresponding to the 111 plane with a d-spacing of 0.87 nm. The observed increase in d-spacing of GO compared with graphite is attributed to the swelling of GO as a result of water absorption as reported in the literature.[15] The nanocomposite GOPS2 shows a major diffraction peak at 18°, which suggests that the nanocomposite has a more ordered structure than the homopolymer. This is also inferred from the uniform size of the particles observed in SEM. The basal spacing of the nanocomposite is also larger than that of graphite but smaller than that of GO, suggesting the formation of intercalated structures. The characteristic diffraction peak of GO at 10° is also present in the nanocomposite showing the presence of both the components in the composite. When a film is prepared from these composites, the PS acts as the matrix in which the GO layers are distributed. In intercalated structures, the polymer chains are present between the platelet layers. The GO layer on the surface of the particles is also seen from the TEM images. (cf. Fig. 5B). Thus, the major peak observed in the XRD pattern of GOPS2 indicates a crystalline microstructure for the nanocomposite, which is different from that of GO and PS, probably as a result of enhanced crystallization.

The major diffraction peak of the nanocomposite corresponds to the 100 plane with an average d-spacing of 0.46 nm and a grain size of 1.75 nm, but for the pure polymer (PS2), the peak was observed for the 220 plane with a d-spacing of 0.49 nm and a grain size of 8 nm. The PS homopolymer without any GO does not show any major diffraction peaks for the 100 plane. The small grain size of the nanocomposite compared with the homopolymer also indicates the presence of smaller sized domains. In comparison, GO possess peaks of the 100 plane with a larger domain size of 4 nm. The average interlayer spacing of 0.46 nm observed in the nanocomposite was different from that of graphite (0.34 nm) or GO (0.87 nm), which suggests a difference in the extent of intercalation.

One of the main objectives in the fabrication of graphene polymer nanocomposites is to obtain a monolayer dispersion

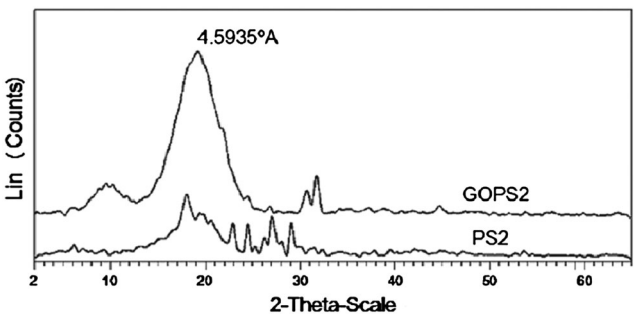

Figure 8. Powder XRD spectra of GOPS2 nanocomposite with 0.5 wt% graphene oxide and PS2 without any graphene oxide.

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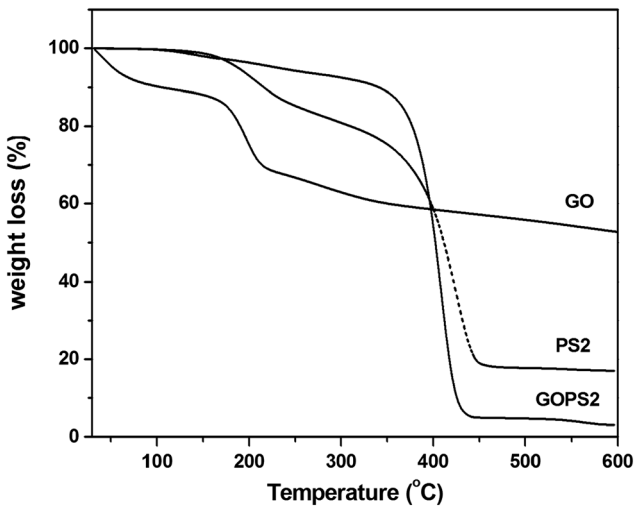

Figure 9. Thermal degradation profile of GOPS2 nanocomposite compared with that of pure polystyrene (PS2) and GO.

of graphene in the composite, which will be reflected in the thermomechanical properties. Thus, thermogravimetric analysis was carried out in order to study the effect of GO on the thermal stability of the nanocomposite. Compared with the homopolymer, the composite shows a marginal decrease of the peak decomposition temperature by about 12°C, but the sample decomposes in a single step at 410°C (Fig. 9). However, the nanocomposite sample displays improved thermal stability features as indicated by a delayed onset of degradation in comparison to pure PS. The thermal degradation of the homopolymer occurs in two steps, first at 206°C (~10% decomposition) and then at 425°C (64% decomposition). The char yield also decreased substantially for the GO-modified sample. The reasons for the reduced char yield may be because of the presence of a large number of oxygen bearing functional groups in GO that promote its oxidation to gaseous by-products like CO and CO_2. Changes in the decomposition pattern of GO–PS nanocomposites have also been observed by other groups [20,25] in agreement with our studies.

CONCLUSIONS

In this work, PS–GO nanocomposites were prepared through seeded emulsion polymerization and characterized for their structure, morphology, optical and thermal properties. The particle size of the nanocomposite dispersions decreases with an increase in GO content. The dispersion on drying displays self-assembly resulting from the uniform size distribution of the particles as observed using SEM, and the dry films exhibit iridescent color as a result of selective reflection of visible light. The reflection wavelength varies with the GO content, and thus, the structural color is tunable. Morphological characterization by SEM and AFM suggests that the shape of the particles changes from spherical to ellipsoid as the GO content is increased from 0.2 to 0.5 wt%. TEM analysis helps to understand the internal structure of the particles. FE-SEM images further confirm the formation of a characteristic crystalline periodic structure on drying. The microstructure of the nanocomposite studied by XRD suggests a characteristic grain size different from that of the pure polymer or GO. The onset temperature of thermal degradation is found to be delayed for the nanocomposite and decomposes in a major step at 410°C. The prepared dispersions may be useful candidates for the study of self-assembly.

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